

Vincent Griffo

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EDUCATION

Georgia Institute of Technology, Atlanta, GA Expected: May 2027
Bachelor of Science in Mechanical Engineering; Minor in Physics; Concentration in Robotics GPA: 4.00/4.00

Harrison High School, Harrison, NY June 2022
High School Diploma: Summa Cum Laude GPA: 4.00/4.00

AWARDS

Tohoku Short-Term Trainee May 2026
Selected from an applicant pool by Tohoku University's School of Engineering to conduct research in Japan under a faculty member of my choice.

Astronaut Scholarship May 2026
National scholarship awarded to top undergraduate researchers demonstrating exceptional academic achievement

Goldwater Scholar March 2026
Competitive merit scholarship awarded to undergraduate students with strong research experience and academics

Georgia Tech Capstone Honorable Mention April 2026
Awarded Honorable Mention for *Sidewalk Sidekick* during the Georgia Tech Spring 2026 Capstone Design Expo, recognizing one of the top 8 non-winning teams among more than 200 capstone teams university-wide.

RSS Space Robotics Workshop Best Paper June 2025
Received Best Paper for GROVER (variable-height grouser wheel) at the RSS Space Robotics Workshop 2025

ThinkSwiss Research Fellowship February 2025
Recipient of a competitive award to research at ETH Zurich under Dr. Marco Hutter of the Robotic Systems Lab

President's Undergraduate Salary Award x2 November 2024, November 2025
Awarded a competitive research grant to support research in the Aerospace Robotics Lab

University Rover Challenge Final Round x2 March 2024, March 2025
Rover advanced to the final round of the University Rover Challenge, competing against 32 international teams at the Mars Desert Research Station in Hanksville, Utah

Stamps President's Scholar August 2022
Full-ride merit scholarship awarded to the top 1% of incoming students at Georgia Tech based on commitment to scholarship, leadership, progress, and service

RESEARCH

Aerospace Robotics Lab | Georgia Tech August 2024 – Present
Principal Investigator: Prof. Yashwanth K. Nakka

Design, Control, and Testing of a Highly Dexterous Multimodal Rover

- Lead a team of seven engineers designing an 8-DOF multimodal rover leg to experimentally study how leg dexterity and morphology influence locomotion efficiency and stability on lunar analog terrain
- Designed low-inertia serial leg architectures using inverse dynamics with MuJoCo and Pinocchio to evaluate joint torque requirements and actuator feasibility across multiple gait morphologies
- Integrated mechanical differentials and transmission strategies to reduce reflected inertia and peak joint torques during multimodal gait transitions

Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals

- Led a multidisciplinary team of six engineers while serving as the project's sole mechanical and controls engineer, developing a 12-DOF (3-DOF/leg) wheeled quadruped capable of 2.7 m/s rolling and 1.05 m/s walking
- Designed the complete mechanical system in SolidWorks, performing inverse dynamics in MuJoCo for actuator sizing and engineering high-load shafts, bearings, transmissions, and structural members validated through finite element analysis on seven machined aluminum components
- Developed the robot's control architecture using rigid-body dynamics, impedance control, dynamic gait generation, CAN bus communication, and EKF-based state estimation for robust multimodal locomotion

- Integrated a Jetson Nano autonomy stack implementing YOLO-based computer vision, obstacle detection, and A* path planning for autonomous navigation and obstacle avoidance
- Manufactured and assembled custom components through CNC machining and waterjet fabrication, culminating in full-system integration and user testing with blind and low-vision participants

Design of Rover Mobility Fleet for Multi-Agent Collaboration (NOVA)

- Led the design of a low-cost collaborative rover fleet for multi-agent autonomy experiments across ARL's lunar terrain testbed to inform future lunar cobot missions
- Designed and manufactured the full rover mechatronic system, including chassis, rocker suspension, and four independent swerve-drive modules enabling omnidirectional motion at speeds up to 2 m/s
- Architected the complete 12 V electrical system, including battery specification, power distribution, actuator selection, relays, and dual Jetson Nano integration
- Integrated dual ZED Mini stereo cameras and onboard perception to enable cooperative navigation and collaborative environment understanding across multiple rovers

Design and Control of Variable-Height Grouser Platform for Extraterrestrial Navigation

- Invented and designed an internal spiral cam mechanism within a wheel that dynamically adjusts grouser height to match analog Martian terrain properties, achieving up to a 58% reduction in slip ratio in granular media and informing how adaptability may aid crucial locomotion in future Martian missions
- Performed structural FEA in SOLIDWORKS on spiral cam and tread geometries, informing stiffness, contact stress, and fatigue limits under dynamic terrain interaction loads
- Designed a fully instrumented terrain testing gantry with linear and rotary encoders, current and voltage sensing to experimentally quantify slip ratio, tractive force, and energy consumption across four terrains
- Developed a closed-loop PID controller in Python to precisely regulate grouser geometry, enabling accurate and repeatable experimental validation of wheel-terrain interaction

Space Infrastructure Engineering Lab | Tohoku University, Japan

May 2026 – July 2026

Principal Investigator: Prof. Toshinori Kuwahara

Design and Control of Electromechanical Soft Docking System for Floating Platform Cubesat

- Designed and integrated a mountable soft-capture docking mechanism with complementary male and female interfaces for autonomous CubeSat rendezvous, capture, and mechanical locking.
- Designed a servo-actuated twist-lock mechanism with ball-detent alignment features, enabling low-friction docking and repeatable locking and undocking operations.
- Designed, wound, and optimized a custom iron-core electromagnet from first principles to enable long-range electromagnetic soft capture on the Space Infrastructure Engineering Lab's air-bearing test platform.
- Applied electromagnetic first principles and a current-controlled test system to maximize magnetic flux while minimizing thermal losses, achieving reliable capture distances of up to 37 cm.

Robotic Systems Lab / LunarLeaper | ETH Zurich, Switzerland

August 2025 – December 2025

Principal Investigator: Prof. Marco Hutter

Design and Control of a 3-DOF Parallel, Space-Grade Arachnid Robot (LunarLeaper)

- Led mission-driven mechanical design of a 3-DOF parallel leg mechanism for *LunarLeaper*, a 12U CubeSat-class quadrupedal robot planned for a 2030 lunar science exploration mission
- Iterated through six full-scale leg prototypes using rapid prototyping and COTS hardware to converge on a compact architecture capable of surviving lunar dust, thermal cycling, and launch constraints
- Executed full mechatronic development of the final prototype using PMSM actuators, employing stationary actuator placement to eliminate moving thermal straps and wiring, improving system lifetime
- Leveraged joint torque and ROM analysis in Simscape Multibody to implement final transmission architectures using heim-joint yaw and hip pitch joints and a steel-shaft flexure knee mechanism

Aerial Robotics and Experimental Autonomy Lab | Georgia Tech

August 2023 – August 2024

Principal Investigator: Prof. Jonathan Rogers

Design and Control of Multimodal UAS Proprotor Articulation for VTOL-to-Fixed-Wing Transition

- Worked on a novel proprotor articulation mechanism to allow seamless transition between vertical take-off and landing (VTOL) and fixed-wing flight modes for a scaled-down unmanned aerial vehicle (UAV)
- Created mathematical models and performed simulations of articulation using MATLAB, applying principles of rigid body mechanics to optimize spatial constraints, maximize torque output, and enhance moment arm rotation
- Designed and implemented a PID control system in MATLAB Simulink, integrating hardware-in-the-loop (HIL) simulation with a DC motor and Arduino Nano for application in the UAS control scheme

Laboratory for Intelligent Decision and Autonomous Robots | Georgia Tech *December 2022 – March 2023*
Principal Investigator: Prof. Ye Zhao

Development of Control Systems and Autonomous Navigation for Human-Avoidance Drones

- Leveraged ROS and Python to design autonomous human-avoidance algorithms to enable real-time drone navigation and obstacle detection for enhanced flight path control

SSP Astrophysics | Online through MIT, Caltech, and Harvey Mudd Sponsorship *May 2021 – July 2021*
Principal Investigators: Prof. Michael Hannawald & Prof. Cassandra Fallscheer

Tracking, Modeling, and Orbital Prediction of Near-Earth Asteroid 2003 HB6

- Determined the position of an asteroid relative to known stars using parallax and coordinate transformations with data from industry-grade telescopes at observatories in Chile, Australia, and the USA
- Calculated orbital parameters of asteroids utilizing the Method of Gauss and Monte Carlo simulation to be used for a precise visual model and accurate approximations within 0.1% of JPL's previously predicted orbital elements
- Developed a mathematical model and 3D visual simulation using Python to predict asteroid positions decades in advance based on determined orbital parameters
- Collaborated with the Southwest Research Institute to analyze asteroid collision history using advanced orbital simulations developed with SWIFT

JOURNAL PUBLICATIONS

- [1] **V. Griffo**, Y. Nakka, Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation (**Submitted IEEE T-FR, revisions requested**).
- [2] **V. Griffo**, B. Ma, J. Yu, Data on Orbit Determination of Asteroid 326732 (2003 HB6) (**Internal to the Harvard Minor Planet Center**).

CONFERENCE PUBLICATIONS

- [1] Y. Nakka, **V. Griffo**, S. Kumar, C. DelGaudio, T. Dokania, A. Bhardwaj, S. Kelkar, U.K. Ponugupati. A Lunar Terrain Facility for Robust and Resilient Learning-Based Autonomy. *IEEE Aerospace Conference*, Big Sky, MT, USA, March 2027. (**Submitted**)
- [2] J. Song, **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, October 2026. (**Accepted abstract**)
- [3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE*, Pittsburgh, PA, USA, October 2026.
- [4] S. Stähler, A. Mittelholz, A. Cocheril, A. Fuhrer, J. Church, E. Krasnova, H. Kolvenbach, J. Smajic, V. Bickel, M. Betschen, E. Brändli, J. Gomez Jodar, **V. Griffo**, L. Heyun, N. Hirzel, Y. Hofmann, J. Isler, Y. Luginbühl, J. Majer, R. Margarit, G. Smith-Vaniz, M. Hutter, R. Kornfeld, LunarLeaper: A university-led legged robotic mission to the moon. *5th Symposium on Space Educational Activities*, Ottobrunn–Taufkirchen, Germany, April 2026.
- [5] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. (**Accepted, Canceled due to Government Shutdown**)
- [6] S. Kumar, **V. Griffo**, Y. Nakka, NOVA: Design Overview of an Autonomous 3D-printed Rover Platform. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. (**Accepted, Canceled due to Government Shutdown**)

[7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025. **(Best Paper)**

ORAL PRESENTATIONS

[1] **V. Griffo**, Adaptive Robotics and Multimodal Exploration, *Astronaut Scholarship Innovators Symposium*, Houston, TX, USA, August 2026.

[2] **V. Griffo**, T. Kuwahara, Design and Control of an Electromechanical Soft-Docking System for Floating-Platform CubeSats. *Tohoku University LQS*, Sendai, Japan, July 2026.

[3] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.

[4] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[5] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025.

[6] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech Annual Research Symposium*, Atlanta, GA, USA, April 2025.

POSTER PRESENTATIONS

[1] J. Song, **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, October 2026. **(Accepted abstract)**

[2] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE*, Pittsburgh, PA, USA, October 2026.

[3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *GT Space Week*, Atlanta, GA, USA, September 2026.

[4] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.

[5] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

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[7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025.

[8] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech College of Sciences EXPLORE*, Atlanta, GA, USA, March 2025.

[9] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Stamps Scholar Summit*, Atlanta, GA, USA, March 2025.

PATENTS

[1] **V. Griffo**, Adaptive Grouser Wheel for Planetary Rovers. *US Provisional Patent Application* No. 63/903,602, filed 22 October 2025. GTRC Ref. 2026-079 (GTRC2026079PRV). United States.

PROFESSIONAL EXPERIENCE

Boston Dynamics, Waltham, MA

July 2026 – Present

ATLAS Systems Test / Mechanical Engineering Co-Op

- Designing and developing custom test rigs to evaluate the dexterous manipulation capabilities of Atlas' end effectors, validating tasks including lifting, reaching, and fine tool manipulation.
- Performing structural analysis and finite element analysis (FEA) to ensure test fixtures exceed required load and stiffness margins, preventing the test apparatus from becoming the failure mode during evaluation.
- Integrating load cells, force/torque sensors, data acquisition systems, and signal conditioning electronics to quantify gripper performance across representative manipulation tasks.
- Designing and validating leg test fixtures to characterize Atlas' lower-limb performance throughout multiple stages of the walking lifecycle using principles of legged robot dynamics.
- Developing mechanical designs, engineering drawings, and production-ready documentation using CATIA V6 while collaborating with multidisciplinary hardware, controls, and test engineering teams.

MIT Lincoln Laboratory, Lexington, MA

May 2024 – August 2024

Interceptor Technology and Systems Architecture Researcher

- Designed and developed a low-cost counter-UAV interceptor, reducing manufacturing time and costs via 3D printing and fastener-free design, set for implementation in operational missions for rapid threat response
- Designed, prototyped, and flight-tested a custom 3D-printed high-torque steering control assembly for a low-cost counter-unmanned aerial system (C-UAS) interceptor missile, enhancing maneuverability and control precision
- Designed and manufactured a deployable fin system utilizing quaternions and eigenvalue decomposition, achieving a 60% reduction in the stowed rocket volume, allowing compact stacking and launching of the interceptors
- Helped develop key subsystems, including PCB avionics, and integrated YOLO-based computer vision for real-time target detection, enhancing autonomous threat response for operational deployment
- Implemented physics-based models using MATLAB and MODTRAN to conduct trade studies of electro-optical/infrared (EO/IR) sensors for unmanned vehicles under adverse atmospheric conditions

Georgia Tech Undergraduate Teaching Assistant, Atlanta, GA

Fall 2023, Fall 2024

Department of Mathematics Linear Algebra Teaching Assistant

- Independently instructed over 40 students in linear algebra, developing comprehensive lecture materials focused on deepening student understanding of key concepts such as eigenvalues, matrix transformations, and vector spaces
- Collaborated with a team of 20+ TAs and professors to coordinate structured grading rubrics and ensure consistent instruction through clear communication and alignment on course objectives and evaluation criteria

COMPETITION TEAMS

URC Mars Rover Team

Member: September 2022 – Present

Former Chief Engineer (two-year term)

- Oversaw the design of an autonomous rover consisting of a rocker-bogie drivetrain, 5-DOF robotic arm, precision end-effector, and carbon fiber chassis for simulated Martian navigation in the University Rover Challenge
- Designed a highly dexterous serial 5-DOF robotic arm using Simscape Multibody analysis, validating rigid-body dynamics and torque requirements for manipulation of a 5 kg payload at 1 m reach
- Designed and fabricated a compliant end-effector to manipulate objects with uncertain geometries, improving operational dexterity across competition tasks
- Led the design and manufacture of a custom carbon fiber drone frame and control scheme, featuring a PID-controlled camera gimbal system integrated with a CUAV X7 flight controller for stabilization and flight control
- Designed and manufactured a 2-DOF robotic arm for the URC drone, enabling manipulation and transport of hand tools for mission-critical operations

Georgia Tech Robotics Competition

August 2023 – November 2023

- Competed against over 200 mechanical engineering students in the Georgia Tech robotics competition, designing and programming an autonomous robot within 5 weeks
- Led the development of a telescoping lift, custom gear drivetrain, high-traction SLA tires, and a passive geometry ball filter, ultimately achieving first seed before the final competition out of 67 teams

VOLUNTEER WORK

Invention Studio Makerspace | *Electronics Apprentice*

September 2022 – March 2024

- Instructed and guided students and makers in the use of various machining tools and techniques, supporting both research and personal projects within the makerspace
- Demonstrated proficiency in circuit design, electromagnetism, microcontrollers, and shop management by creating complex electronic projects, such as a Gauss Accelerator, and managing the budget to order and restock makerspace components

Skills

- **Technical Software:** SOLIDWORKS, SOLIDWORKS FEA, Siemens NX, Fusion 360, ANSYS, MATLAB, Simulink, Simscape Multibody, MuJoCo, Mission Planner, PX4, Jetson SDK, Linux
- **Hardware & Prototyping:** Precision machining (lathe, mill, waterjet, PCB mill), additive manufacturing (FDM/SLA), TIG/MIG welding, soldering and wiring, sensor integration, rapid prototyping, electromechanical assembly
- **Programming & Embedded Systems:** C++, Python, MATLAB, C#, Java, JavaScript, ROS, Arduino IDE, Raspberry Pi OS, $\text{\LaTeX}$
- **Robotics & Control:** Robot kinematics and inverse dynamics, Jacobian-based analysis, legged locomotion, state estimation, actuator design and sizing, PID and field-oriented control (FOC), embedded serial interfaces, hardware-in-the-loop (HIL) control, data acquisition